

# LM386 Low Voltage Audio Power Amplifier

## 1 Features

- Battery Operation
- Minimum External Parts
- Wide Supply Voltage Range: 4 V–12 V or 5 V–18 V
- Low Quiescent Current Drain: 4 mA
- Voltage Gains from 20 to 200
- Ground-Referenced Input
- Self-Centering Output Quiescent Voltage
- Low Distortion: 0.2% ( $A_V = 20$ ,  $V_S = 6$  V,  $R_L = 8$   $\Omega$ ,  $P_O = 125$  mW,  $f = 1$  kHz)
- Available in 8-Pin MSOP Package

## 2 Applications

- AM-FM Radio Amplifiers
- Portable Tape Player Amplifiers
- Intercoms
- TV Sound Systems
- Line Drivers
- Ultrasonic Drivers
- Small Servo Drivers
- Power Converters

## 3 Description

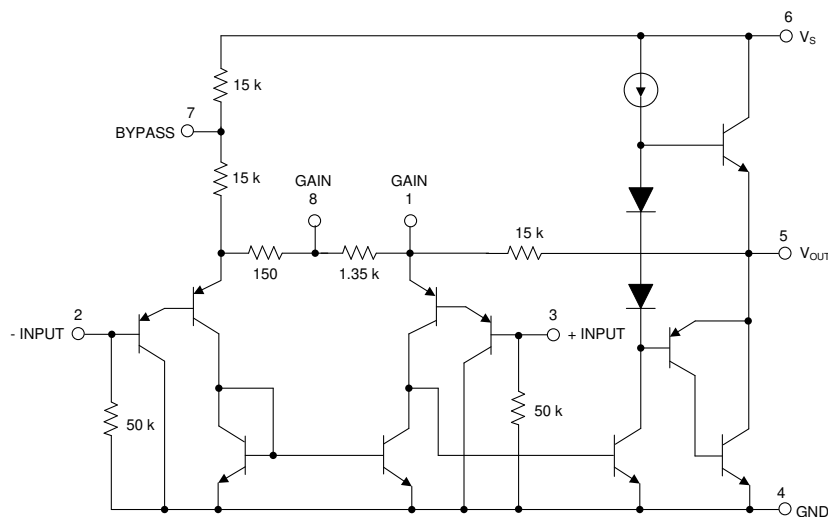
The LM386M-1 and LM386MX-1 are power amplifiers designed for use in low voltage consumer applications. The gain is internally set to 20 to keep external part count low, but the addition of an external resistor and capacitor between pins 1 and 8 increases the gain to any value from 20 to 200.

The inputs are ground referenced while the output automatically biases to one-half the supply voltage. The quiescent power drain is only 24 mW when operating from a 6-V supply, making the LM386M-1 and LM386MX-1 preferred for battery operation.

### Device Information<sup>(1)</sup>

| PART NUMBER | PACKAGE   | BODY SIZE (NOM)   |
|-------------|-----------|-------------------|
| LM386N-1    | PDIP (8)  | 9.60 mm × 6.35 mm |
| LM386N-3    | PDIP (8)  | 9.60 mm × 6.35 mm |
| LM386N-4    | PDIP (8)  | 9.60 mm × 6.35 mm |
| LM386M-1    | SOIC (8)  | 4.90 mm × 3.90 mm |
| LM386MX-1   | SOIC (8)  | 4.90 mm × 3.90 mm |
| LM386MMX-1  | VSSOP (8) | 3.00 mm × 3.00 mm |

(1) For all available packages, see the orderable addendum at the end of the data sheet.



Schematic



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## 4 Revision History

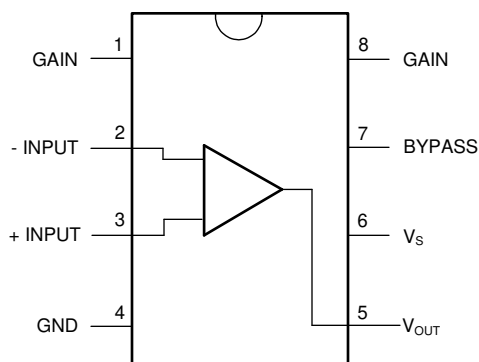
NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

| <b>Changes from Revision C (May 2017) to Revision D (August 2023)</b> | <b>Page</b> |
|-----------------------------------------------------------------------|-------------|
| • Updated Typical Output Power Spec.....                              | <b>5</b>    |

| <b>Changes from Revision B (March 2017) to Revision C (May 2017)</b>                                            | <b>Page</b> |
|-----------------------------------------------------------------------------------------------------------------|-------------|
| • Changed devices LM386M-1/LM386MX-1 To: LM386 in the data sheet title.....                                     | <b>1</b>    |
| • Updated the numbering format for tables, figures, and cross-references throughout the document.....           | <b>1</b>    |
| • Changed From: LM386N-4 To: Speaker Impedance in the <i>Recommended Operating Conditions</i> table.....        | <b>4</b>    |
| • Changed From: 5 $\Omega$ to 12 $\Omega$ To: 5 V to 12 V for Supply Voltage in <a href="#">Table 9-1</a> ..... | <b>10</b>   |
| • Changed kW To: k $\Omega$ in the <i>Gain Control</i> section.....                                             | <b>10</b>   |
| • Changed kW To: k $\Omega$ in the <i>Input Biasing</i> section.....                                            | <b>11</b>   |
| • Changed <a href="#">Figure 9-2</a> .....                                                                      | <b>11</b>   |
| • Changed From: 5 $\Omega$ to 12 $\Omega$ To: 5 V to 12 V for Supply Voltage in <a href="#">Table 9-2</a> ..... | <b>12</b>   |
| • Changed <a href="#">Figure 9-4</a> .....                                                                      | <b>12</b>   |
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| • Changed <a href="#">Figure 9-6</a> .....                                                                      | <b>13</b>   |
| • Changed From: 5 $\Omega$ to 12 $\Omega$ To: 5 V to 12 V for Supply Voltage in <a href="#">Table 9-4</a> ..... | <b>14</b>   |
| • Changed <a href="#">Figure 9-8</a> .....                                                                      | <b>14</b>   |
| • Changed From: 5 $\Omega$ to 12 $\Omega$ To: 5 V to 12 V for Supply Voltage in <a href="#">Table 9-5</a> ..... | <b>15</b>   |
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| • Changed <a href="#">Figure 9-12</a> .....                                                                     | <b>16</b>   |
| • Changed From: 5 $\Omega$ to 12 $\Omega$ To: 5 V to 12 V for Supply Voltage in <a href="#">Table 9-7</a> ..... | <b>17</b>   |
| • Changed <a href="#">Figure 9-14</a> .....                                                                     | <b>17</b>   |

| <b>Changes from Revision A (May 2004) to Revision B (March 2017)</b>                                                                                | <b>Page</b> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| • Added LM386MX-1 device to the data sheet. ....                                                                                                    | <b>1</b>    |
| • Added Device Information, Application and Implementation, Power Supply Recommendation, Layout, and Device and Documentation Support sections..... | <b>1</b>    |
| • Inserted Functional Block Diagram.....                                                                                                            | <b>9</b>    |

## 5 Pin Configuration and Functions



**Figure 5-1. D Package 8-Pin MSOP Top View**

**Table 5-1. Pin Functions**

| PIN              |     | TYPE <sup>(1)</sup> | DESCRIPTION            |
|------------------|-----|---------------------|------------------------|
| NAME             | NO. |                     |                        |
| GAIN             | 1   | –                   | Gain setting pin       |
| –INPUT           | 2   | I                   | Inverting input        |
| +INPUT           | 3   | I                   | Noninverting input     |
| GND              | 4   | P                   | Ground reference       |
| V <sub>OUT</sub> | 5   | O                   | Output                 |
| V <sub>S</sub>   | 6   | P                   | Power supply voltage   |
| BYPASS           | 7   | O                   | Bypass decoupling path |
| GAIN             | 8   | –                   | Gain setting pin       |

(1) I = Input, O = Output, P = Power

## 6 Specifications

### 6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)<sup>(1)</sup>

|                                |                       | MIN  | MAX   | UNIT |
|--------------------------------|-----------------------|------|-------|------|
| Supply Voltage, $V_{CC}$       | LM386N-1/-3, LM386M-1 |      | 15    | V    |
|                                | LM386N-4              |      | 22    |      |
| Package Dissipation            | LM386N                |      | 1.25  | W    |
|                                | LM386M                |      | 0.73  |      |
|                                | LM386MM-1             |      | 0.595 |      |
| Input Voltage, $V_I$           |                       | −0.4 | 0.4   | V    |
| Storage temperature, $T_{stg}$ |                       | −65  | 150   | °C   |

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

### 6.2 ESD Ratings

|                                     |                                                                                | VALUE | UNIT |
|-------------------------------------|--------------------------------------------------------------------------------|-------|------|
| $V_{(ESD)}$ Electrostatic discharge | Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 <sup>(1)</sup>              | ±1000 | V    |
|                                     | Charged-device model (CDM), per JEDEC specification JESD22-C101 <sup>(2)</sup> | ±1000 |      |

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

### 6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

|          |                                | MIN  | NOM | MAX | UNIT |
|----------|--------------------------------|------|-----|-----|------|
| $V_{CC}$ | Supply Voltage                 | 4    |     | 12  | V    |
|          | LM386N-4                       | 5    |     | 18  | V    |
|          | Speaker Impedance              | 4    |     |     | Ω    |
| $V_I$    | Analog input voltage           | −0.4 |     | 0.4 | V    |
| $T_A$    | Operating free-air temperature | 0    |     | 70  | °C   |

### 6.4 Thermal Information

| THERMAL METRIC <sup>(1)</sup> |                                              | LM386    | LM386       | LM386    | UNIT |
|-------------------------------|----------------------------------------------|----------|-------------|----------|------|
|                               |                                              | D (SOIC) | DGK (VSSOP) | P (PDIP) |      |
|                               |                                              | 8        | 8           | 8        |      |
| $R_{\theta JA}$               | Junction-to-ambient thermal resistance       | 115.7    | 169.3       | 53.4     | °C/W |
| $R_{\theta JC(top)}$          | Junction-to-case (top) thermal resistance    | 59.7     | 73.1        | 42.1     | °C/W |
| $R_{\theta JB}$               | Junction-to-board thermal resistance         | 56.2     | 100.2       | 30.6     | °C/W |
| $\Psi_{JT}$                   | Junction-to-top characterization parameter   | 12.4     | 9.2         | 19.0     | °C/W |
| $\Psi_{JB}$                   | Junction-to-board characterization parameter | 55.6     | 99.1        | 50.5     | °C/W |

- (1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

## 6.5 Electrical Characteristics

over operating free-air temperature range (unless otherwise noted)

| PARAMETER  |                              | TEST CONDITIONS                                                                                               | MIN | TYP  | MAX | UNIT       |
|------------|------------------------------|---------------------------------------------------------------------------------------------------------------|-----|------|-----|------------|
| $V_S$      | Operating Supply Voltage     | LM386N-1, -3, LM386M-1, LM386MM-1                                                                             | 4   |      | 12  | V          |
|            |                              | LM386N-4                                                                                                      | 5   |      | 18  |            |
| $I_Q$      | Quiescent Current            | $V_S = 6\text{ V}$ , $V_{IN} = 0$                                                                             |     | 4    | 8   | mA         |
| $P_{OUT}$  | Output Power                 | $V_S = 6\text{ V}$ , $R_L = 8\ \Omega$ , THD = 10%<br>(LM386N-1, LM386M-1, LM386MM-1)                         | 250 | 325  |     | mW         |
|            |                              | $V_S = 9\text{ V}$ , $R_L = 8\ \Omega$ , THD = 10%<br>(LM386N-3)                                              | 500 | 700  |     |            |
|            |                              | $V_S = 16\text{ V}$ , $R_L = 32\ \Omega$ , THD = 10%<br>(LM386N-4)                                            | 700 | 1000 |     |            |
| $A_V$      | Voltage Gain                 | $V_S = 6\text{ V}$ , $f = 1\text{ kHz}$                                                                       |     | 26   |     | dB         |
|            |                              | 10 $\mu\text{F}$ from Pin 1 to 8                                                                              |     | 46   |     |            |
| BW         | Bandwidth                    | $V_S = 6\text{ V}$ , Pins 1 and 8 Open                                                                        |     | 300  |     | kHz        |
| THD        | Total Harmonic Distortion    | $V_S = 6\text{ V}$ , $R_L = 8\ \Omega$ , $P_{OUT} = 125\text{ mW}$<br>$f = 1\text{ kHz}$ , Pins 1 and 8 Open  |     | 0.2% |     |            |
| PSRR       | Power Supply Rejection Ratio | $V_S = 6\text{ V}$ , $f = 1\text{ kHz}$ , CBYPASS = 10 $\mu\text{F}$<br>Pins 1 and 8 Open, Referred to Output |     | 50   |     | dB         |
| $R_{IN}$   | Input Resistance             |                                                                                                               |     | 50   |     | k $\Omega$ |
| $I_{BIAS}$ | Input Bias Current           | $V_S = 6\text{ V}$ , Pins 2 and 3 Open                                                                        |     | 250  |     | nA         |

## 6.6 Typical Characteristics

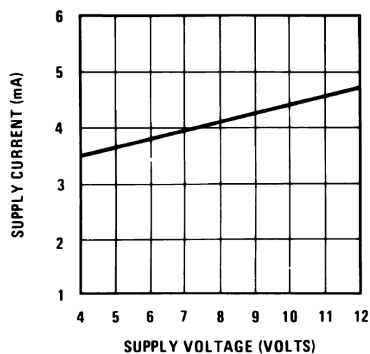


Figure 6-1. Supply Current vs Supply Voltage

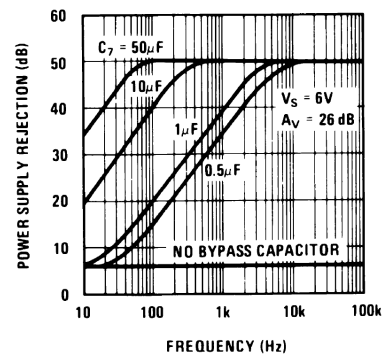


Figure 6-2. Power Supply Rejection vs Frequency

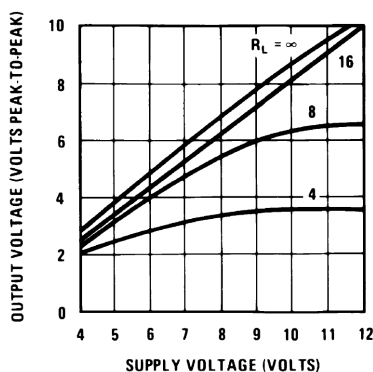


Figure 6-3. Output Voltage vs Supply Voltage

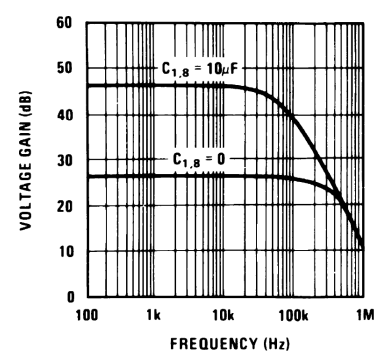


Figure 6-4. Voltage Gain vs Frequency

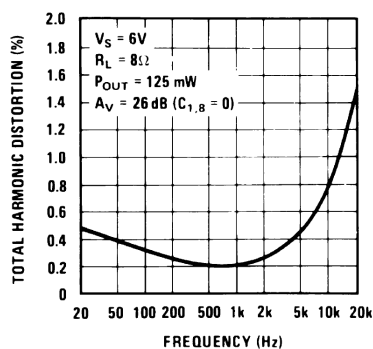


Figure 6-5. Total Harmonic Distortion vs Frequency

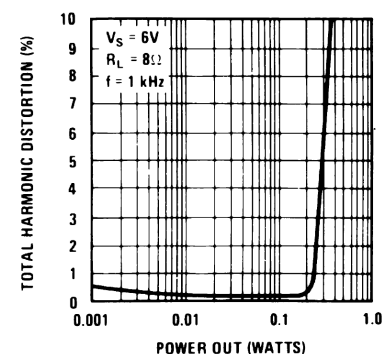


Figure 6-6. Total Harmonic Distortion vs Power Out

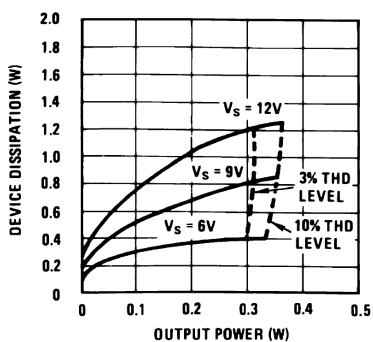


Figure 6-7. Device Dissipation vs Output Power

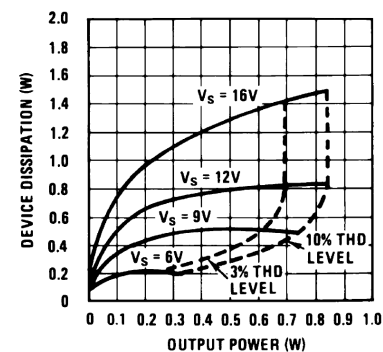
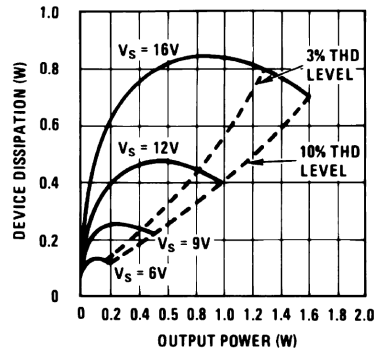


Figure 6-8. Device Dissipation vs Output Power



**Figure 6-9. Device Dissipation vs Output Power**

## 7 Parameter Measurement Information

All parameters are measured according to the conditions described in the [Section 6](#) section.

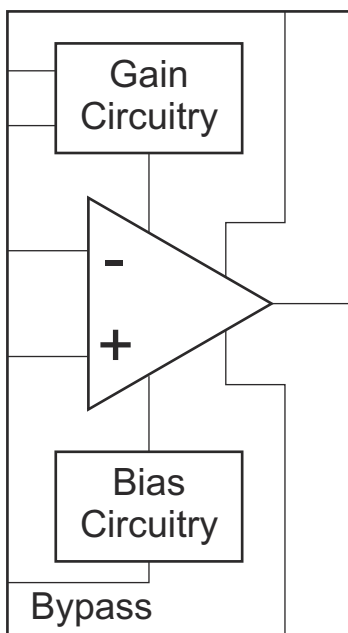


## 8 Detailed Description

### 8.1 Overview

The LM386 is a mono low voltage amplifier that can be used in a variety of applications. It can drive loads from 4  $\Omega$  to 32  $\Omega$ . The gain is internally set to 20 but it can be modified from 20 to 200 by placing a resistor and capacitor between pins 1 and 8. This device comes in three different 8-pin packages as PDIP, SOIC and VSSOP to fit in different applications.

### 8.2 Functional Block Diagram



### 8.3 Feature Description

There is an internal 1.35-K $\Omega$  resistor that sets the gain of this device to 20. The gain can be modified from 20 to 200. Detailed information about gain setting can be found in the [Section 9.2.2.2](#) section.

### 8.4 Device Functional Modes

As this is an Op Amp it can be used in different configurations to fit in several applications. The internal gain setting resistor allows the LM386 to be used in a very low part count system. In addition a series resistor can be placed between pins 1 and 5 to modify the gain and frequency response for specific applications.

## 9 Application and Implementation

### Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

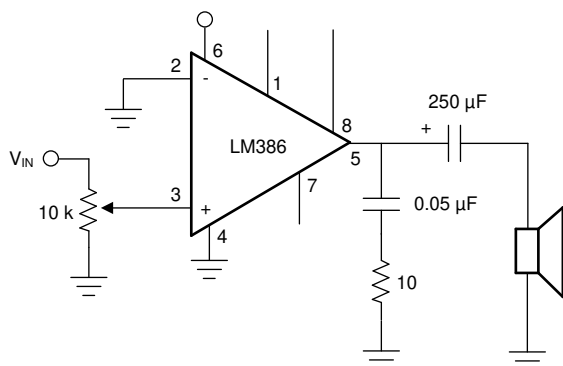
### 9.1 Application Information

Below are shown different setups that show how the LM386 can be implemented in a variety of applications.

### 9.2 Typical Application

#### 9.2.1 LM386 with Gain = 20

Figure 9-1 shows the minimum part count application that can be implemented using LM386. Its gain is internally set to 20.



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**Figure 9-1. LM386 with Gain = 20**

#### 9.2.1.1 Design Requirements

**Table 9-1. Design Parameters**

| DESIGN PARAMETER | EXAMPLE VALUE |
|------------------|---------------|
| Load Impedance   | 4 Ω to 32 Ω   |
| Supply Voltage   | 5 V to 12 V   |

#### 9.2.1.2 Detailed Design Procedure

##### 9.2.1.2.1 Gain Control

To make the LM386 a more versatile amplifier, two pins (1 and 8) are provided for gain control. With pins 1 and 8 open the 1.35-kΩ resistor sets the gain at 20 (26 dB). If a capacitor is put from pin 1 to 8, bypassing the 1.35-kΩ resistor, the gain will go up to 200 (46 dB). If a resistor is placed in series with the capacitor, the gain can be set to any value from 20 to 200. Gain control can also be done by capacitively coupling a resistor (or FET) from pin 1 to ground.

Additional external components can be placed in parallel with the internal feedback resistors to tailor the gain and frequency response for individual applications. For example, we can compensate poor speaker bass response by frequency shaping the feedback path. This is done with a series RC from pin 1 to 5 (paralleling the internal

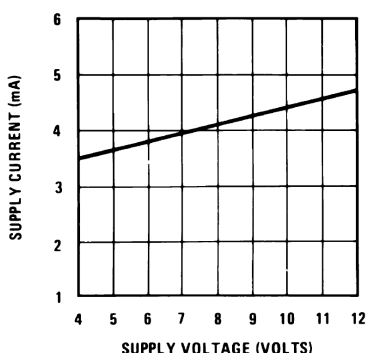
15-kΩ resistor). For 6 dB effective bass boost:  $R \approx 15 \text{ k}\Omega$ , the lowest value for good stable operation is  $R = 10 \text{ k}\Omega$  if pin 8 is open. If pins 1 and 8 are bypassed then  $R$  as low as 2 kΩ can be used. This restriction is because the amplifier is only compensated for closed-loop gains greater than 9.

#### 9.2.1.2.2 Input Biasing

The schematic shows that both inputs are biased to ground with a 50 k $\Omega$  resistor. The base current of the input transistors is about 250 nA, so the inputs are at about 12.5 mV when left open. If the dc source resistance driving the LM386 is higher than 250 k $\Omega$  it will contribute very little additional offset (about 2.5 mV at the input, 50 mV at the output). If the dc source resistance is less than 10 k $\Omega$ , then shorting the unused input to ground will keep the offset low (about 2.5 mV at the input, 50 mV at the output). For dc source resistances between these values we can eliminate excess offset by putting a resistor from the unused input to ground, equal in value to the dc source resistance. Of course all offset problems are eliminated if the input is capacitively coupled.

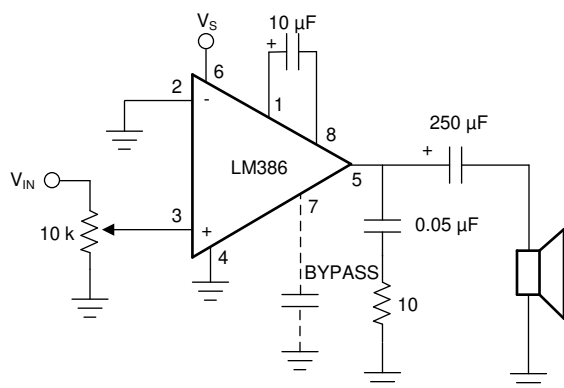
When using the LM386 with higher gains (bypassing the 1.35 k $\Omega$  resistor between pins 1 and 8) it is necessary to bypass the unused input, preventing degradation of gain and possible instabilities. This is done with a 0.1  $\mu$ F capacitor or a short to ground depending on the dc source resistance on the driven input.

#### 9.2.1.3 Application Curve



**Figure 9-2. Supply Current vs Supply Voltage**

## 9.2.2 LM386 with Gain = 200



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**Figure 9-3. LM386 with Gain = 200**

### 9.2.2.1 Design Requirements

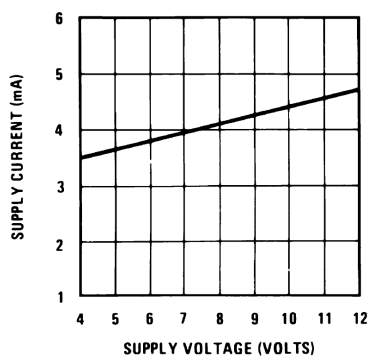
**Table 9-2. Design Parameters**

| DESIGN PARAMETER | EXAMPLE VALUE |
|------------------|---------------|
| Load Impedance   | 4 Ω to 32 Ω   |
| Supply Voltage   | 5 V to 12 V   |

### 9.2.2.2 Detailed Design Procedure

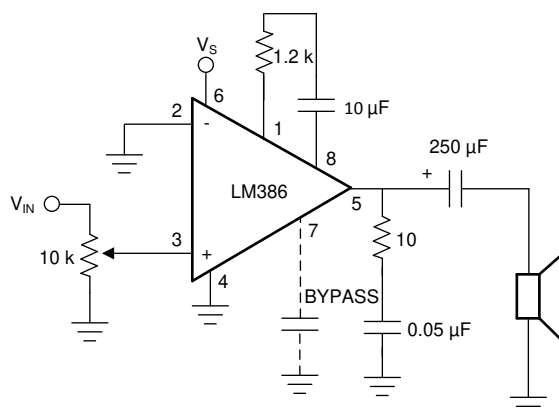
The Detailed Design Procedure can be found in the [Section 9.2.1.2](#) section.

### 9.2.2.3 Application Curve



**Figure 9-4. Supply Current vs Supply Voltage**

### 9.2.3 LM386 with Gain = 50



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**Figure 9-5. LM386 with Gain = 50**

#### 9.2.3.1 Design Requirements

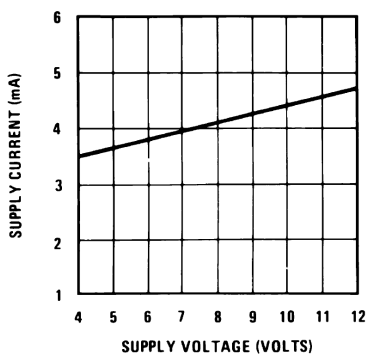
**Table 9-3. Design Parameters**

| DESIGN PARAMETER | EXAMPLE VALUE |
|------------------|---------------|
| Load Impedance   | 4 Ω to 32 Ω   |
| Supply Voltage   | 5 V to 12 V   |

#### 9.2.3.2 Detailed Design Procedure

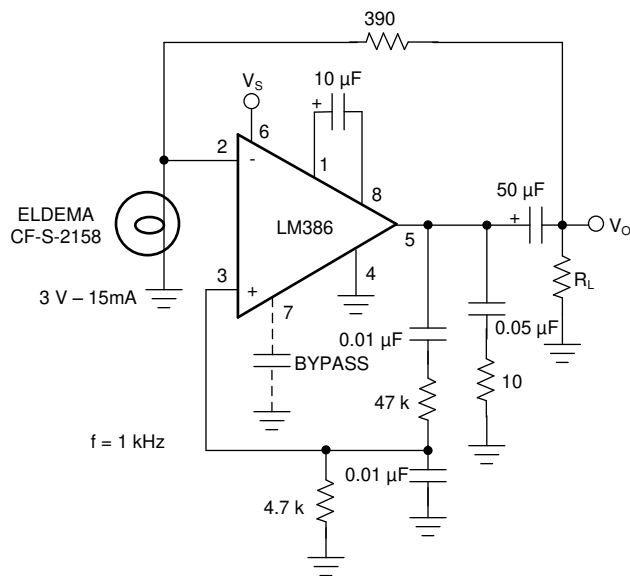
The Detailed Design Procedure can be found in the [Section 9.2.1.2](#) section.

#### 9.2.3.3 Application Curve



**Figure 9-6. Supply Current vs Supply Voltage**

## 9.2.4 Low Distortion Power Wienbridge Oscillator



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**Figure 9-7. Low Distortion Power Wienbridge Oscillator**

### 9.2.4.1 Design Requirements

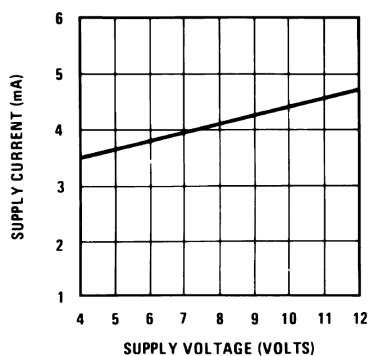
**Table 9-4. Design Parameters**

| DESIGN PARAMETER | EXAMPLE VALUE |
|------------------|---------------|
| Load Impedance   | 4 Ω to 32 Ω   |
| Supply Voltage   | 5 V to 12 V   |

### 9.2.4.2 Detailed Design Procedure

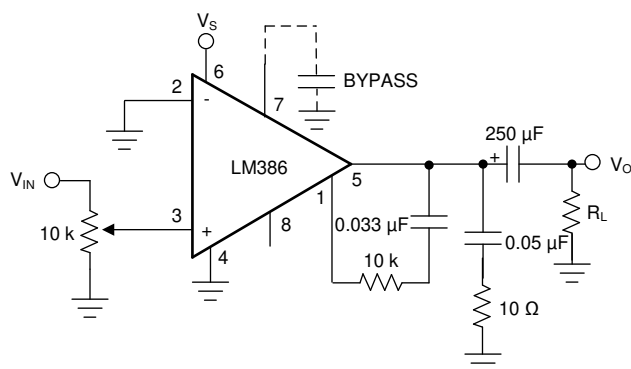
The Detailed Design Procedure can be found in the [Section 9.2.1.2](#) section.

### 9.2.4.3 Application Curve



**Figure 9-8. Supply Current vs Supply Voltage**

## 9.2.5 LM386 with Bass Boost



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**Figure 9-9. LM386 with Bass Boost**

### 9.2.5.1 Design Requirements

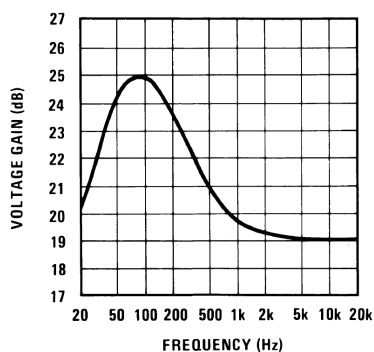
**Table 9-5. Design Parameters**

| DESIGN PARAMETER | EXAMPLE VALUE |
|------------------|---------------|
| Load Impedance   | 4 Ω to 32 Ω   |
| Supply Voltage   | 5 V to 12 V   |

### 9.2.5.2 Detailed Design Procedure

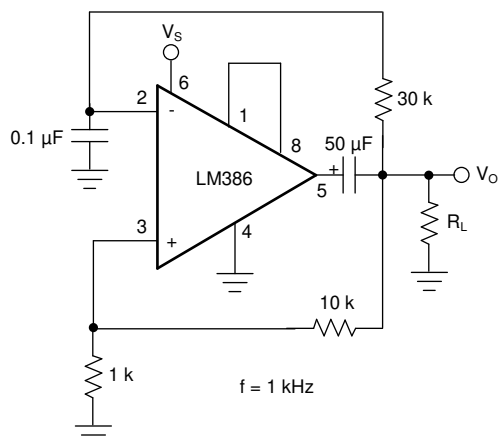
The Detailed Design Procedure can be found in the [Section 9.2.1.2](#) section.

### 9.2.5.3 Application Curve



**Figure 9-10. Voltage Gain vs Frequency**

## 9.2.6 Square Wave Oscillator



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**Figure 9-11. Square Wave Oscillator**

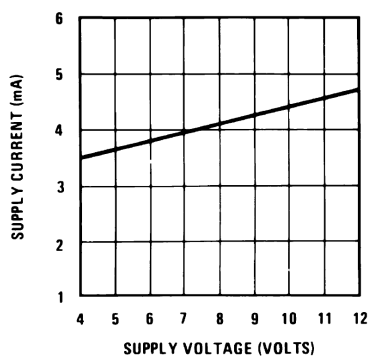
**Table 9-6. Design Parameters**

| DESIGN PARAMETER | EXAMPLE VALUE |
|------------------|---------------|
| Load Impedance   | 4 Ω to 32 Ω   |
| Supply Voltage   | 5 V to 12 V   |

### 9.2.6.1 Detailed Design Procedure

The Detailed Design Procedure can be found in the [Section 9.2.1.2](#) section.

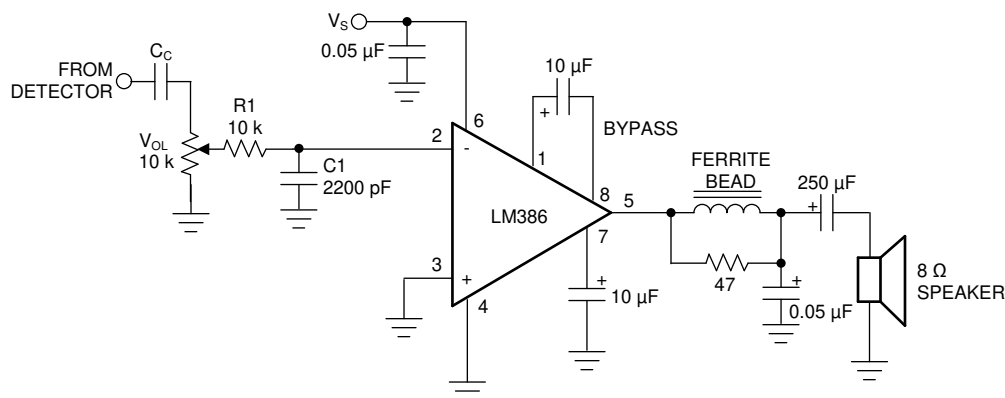
### 9.2.6.2 Application Curve



**Figure 9-12. Supply Current vs Supply Voltage**



## 9.2.7 AM Radio Power Amplifier



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**Figure 9-13. AM Radio Power Amplifier**

### 9.2.7.1 Design Requirements

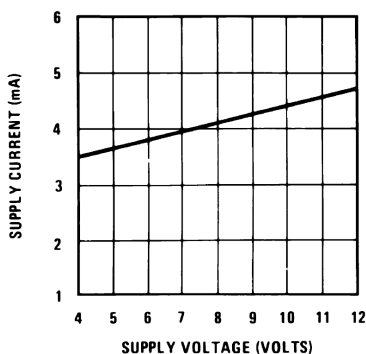
**Table 9-7. Design Parameters**

| DESIGN PARAMETER | EXAMPLE VALUE             |
|------------------|---------------------------|
| Load Impedance   | 4 $\Omega$ to 32 $\Omega$ |
| Supply Voltage   | 5 V to 12 V               |

### 9.2.7.2 Detailed Design Procedure

The Detailed Design Procedure can be found in the [Section 9.2.1.2](#) section.

### 9.2.7.3 Application Curve



**Figure 9-14. Supply Current vs Supply Voltage**

## 10 Power Supply Recommendations

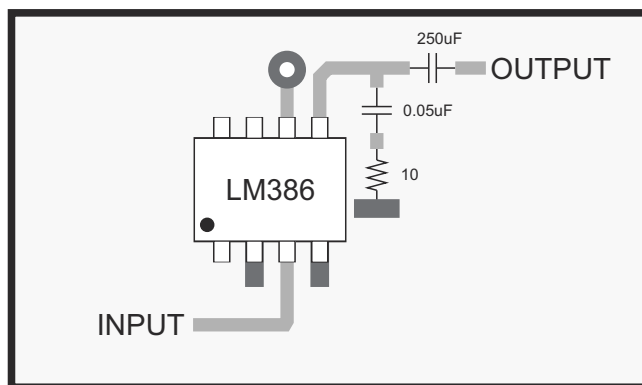
The LM386 is specified for operation up to 12 V or 18 V. The power supply should be well regulated and the voltage must be within the specified values. It is recommended to place a capacitor to GND close to the LM386 power supply pin.

## 11 Layout

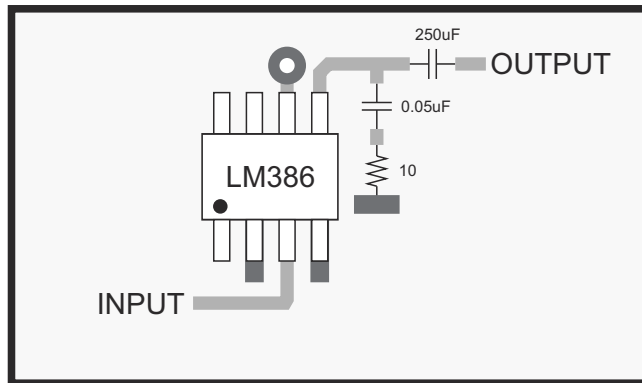
### 11.1 Layout Guidelines

Place all required components as close as possible to the device. Use short traces for the output to the speaker connection. Route the analog traces far from the digital signal traces and avoid crossing them.

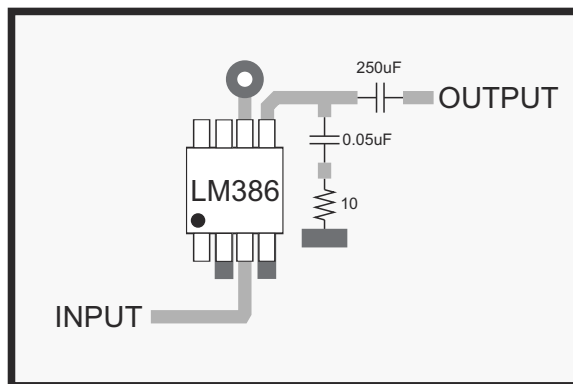
### 11.2 Layout Examples



**Figure 11-1. Layout Example for Minimum Parts Gain = 20 dB on PDIP package**



**Figure 11-2. Layout Example for Minimum Parts Gain = 20 dB on SOIC package**



**Figure 11-3. Layout Example for Minimum Parts Gain = 20 dB on VSSOP package**

## 12 Device and Documentation Support

### 12.1 Device Support

#### 12.1.1 Development Support

### 12.2 Documentation Support

### 12.3 Receiving Notification of Documentation Updates

To receive notification of documentation updates — go to the product folder for your device on ti.com. In the upper right-hand corner, click the *Alert me* button to register and receive a weekly digest of product information that has changed (if any). For change details, check the revision history of any revised document.

### 12.4 Community Resources

### 12.5 Trademarks

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## Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

## PACKAGING INFORMATION

| Orderable part number           | Status<br>(1) | Material type<br>(2) | Package   Pins  | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6) |
|---------------------------------|---------------|----------------------|-----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|---------------------|
| <a href="#">LM386M-1/NOPB</a>   | Active        | Production           | SOIC (D)   8    | 95   TUBE             | Yes         | SN                                   | Level-1-260C-UNLIM                | 0 to 70      | LM386<br>M-1        |
| LM386M-1/NOPB.B                 | Active        | Production           | SOIC (D)   8    | 95   TUBE             | Yes         | SN                                   | Level-1-260C-UNLIM                | 0 to 70      | LM386<br>M-1        |
| <a href="#">LM386MMX-1/NOPB</a> | Active        | Production           | VSSOP (DGK)   8 | 3500   LARGE T&R      | Yes         | SN                                   | Level-1-260C-UNLIM                | 0 to 70      | Z86                 |
| LM386MMX-1/NOPB.B               | Active        | Production           | VSSOP (DGK)   8 | 3500   LARGE T&R      | Yes         | SN                                   | Level-1-260C-UNLIM                | 0 to 70      | Z86                 |
| <a href="#">LM386MX-1/NOPB</a>  | Active        | Production           | SOIC (D)   8    | 2500   LARGE T&R      | Yes         | SN                                   | Level-1-260C-UNLIM                | 0 to 70      | LM386<br>M-1        |
| LM386MX-1/NOPB.B                | Active        | Production           | SOIC (D)   8    | 2500   LARGE T&R      | Yes         | SN                                   | Level-1-260C-UNLIM                | 0 to 70      | LM386<br>M-1        |
| <a href="#">LM386N-1/NOPB</a>   | Active        | Production           | PDIP (P)   8    | 40   TUBE             | Yes         | NIPDAU                               | Level-1-NA-UNLIM                  | 0 to 70      | LM<br>386N-1        |
| LM386N-1/NOPB.B                 | Active        | Production           | PDIP (P)   8    | 40   TUBE             | Yes         | NIPDAU                               | Level-1-NA-UNLIM                  | 0 to 70      | LM<br>386N-1        |
| <a href="#">LM386N-3/NOPB</a>   | Active        | Production           | PDIP (P)   8    | 40   TUBE             | Yes         | NIPDAU                               | Level-1-NA-UNLIM                  | 0 to 70      | LM<br>386N-3        |
| LM386N-3/NOPB.B                 | Active        | Production           | PDIP (P)   8    | 40   TUBE             | Yes         | NIPDAU                               | Level-1-NA-UNLIM                  | 0 to 70      | LM<br>386N-3        |
| LM386N-3/NOPBG4                 | Active        | Production           | PDIP (P)   8    | 40   TUBE             | Yes         | NIPDAU                               | Level-1-NA-UNLIM                  | 0 to 70      | LM<br>386N-3        |
| LM386N-3/NOPBG4.B               | Active        | Production           | PDIP (P)   8    | 40   TUBE             | Yes         | NIPDAU                               | Level-1-NA-UNLIM                  | 0 to 70      | LM<br>386N-3        |
| <a href="#">LM386N-4/NOPB</a>   | Active        | Production           | PDIP (P)   8    | 40   TUBE             | Yes         | NIPDAU                               | Level-1-NA-UNLIM                  | 0 to 70      | LM<br>386N-4        |
| LM386N-4/NOPB.B                 | Active        | Production           | PDIP (P)   8    | 40   TUBE             | Yes         | NIPDAU                               | Level-1-NA-UNLIM                  | 0 to 70      | LM<br>386N-4        |
| LM386N-4/NOPBG4                 | Active        | Production           | PDIP (P)   8    | 40   TUBE             | Yes         | NIPDAU                               | Level-1-NA-UNLIM                  | 0 to 70      | LM<br>386N-4        |
| LM386N-4/NOPBG4.B               | Active        | Production           | PDIP (P)   8    | 40   TUBE             | Yes         | NIPDAU                               | Level-1-NA-UNLIM                  | 0 to 70      | LM<br>386N-4        |

<sup>(1)</sup> **Status:** For more details on status, see our [product life cycle](#).

<sup>(2)</sup> **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

<sup>(3)</sup> **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

<sup>(4)</sup> **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

<sup>(5)</sup> **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

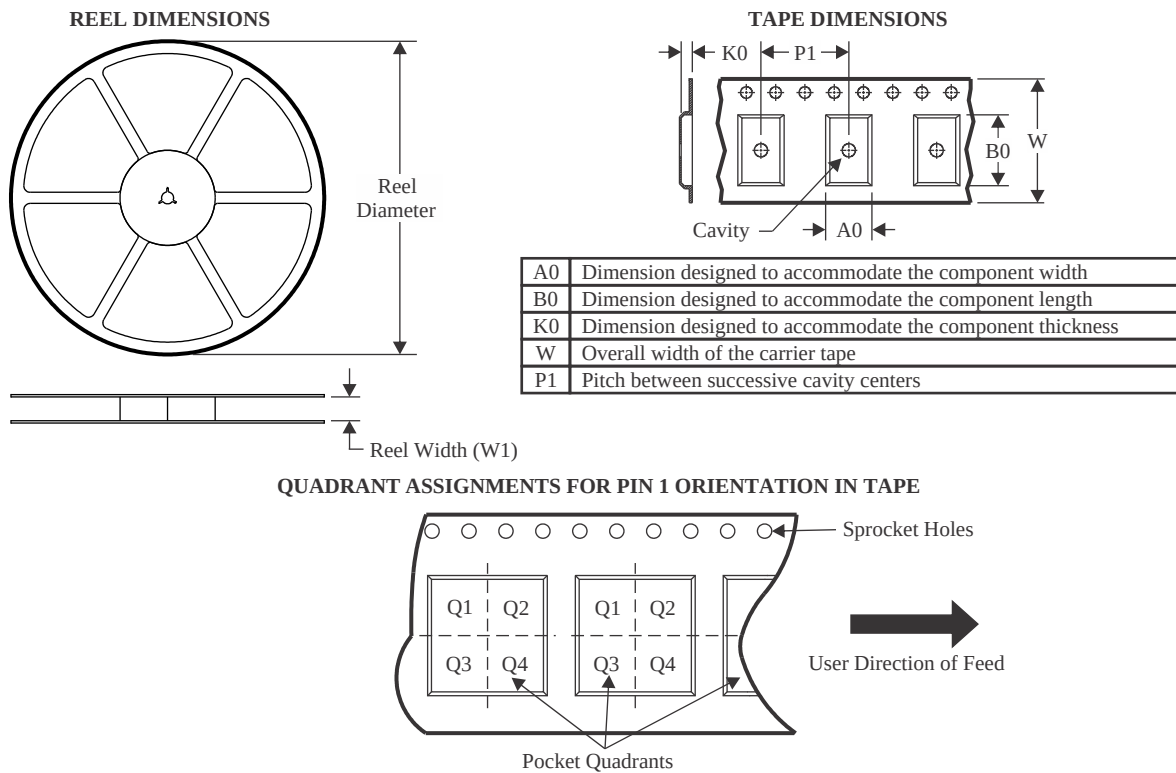
<sup>(6)</sup> **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

## TAPE AND REEL INFORMATION

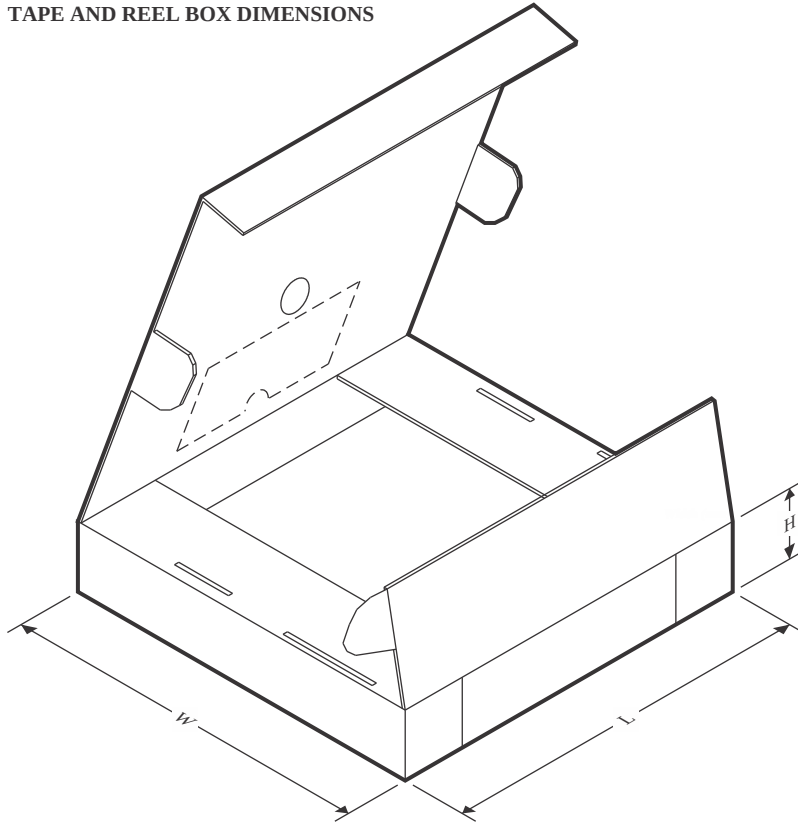


\*All dimensions are nominal

| Device          | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|-----------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| LM386MMX-1/NOPB | VSSOP        | DGK             | 8    | 3500 | 330.0              | 12.4               | 5.3     | 3.4     | 1.4     | 8.0     | 12.0   | Q1            |
| LM386MX-1/NOPB  | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.5     | 5.4     | 2.0     | 8.0     | 12.0   | Q1            |

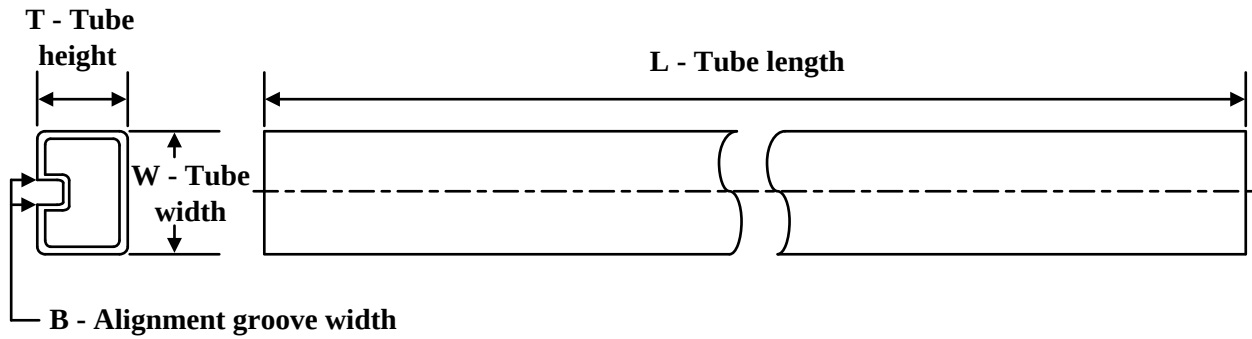


## TAPE AND REEL BOX DIMENSIONS



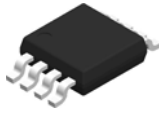
\*All dimensions are nominal

| Device          | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|-----------------|--------------|-----------------|------|------|-------------|------------|-------------|
| LM386MMX-1/NOPB | VSSOP        | DGK             | 8    | 3500 | 367.0       | 367.0      | 35.0        |
| LM386MX-1/NOPB  | SOIC         | D               | 8    | 2500 | 367.0       | 367.0      | 35.0        |

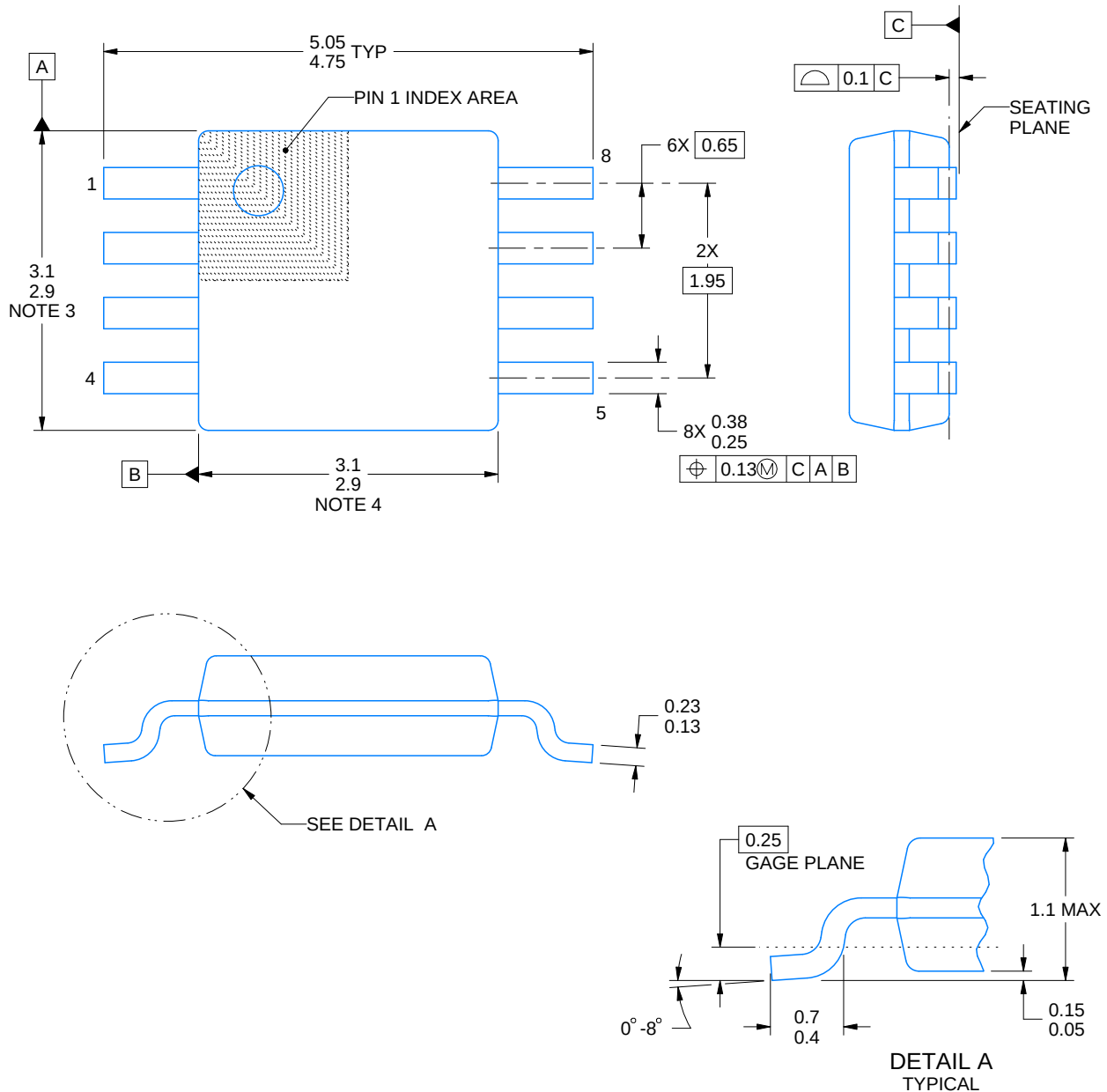
**TUBE**


\*All dimensions are nominal

| Device            | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (μm) | B (mm) |
|-------------------|--------------|--------------|------|-----|--------|--------|--------|--------|
| LM386M-1/NOPB     | D            | SOIC         | 8    | 95  | 495    | 8      | 4064   | 3.05   |
| LM386M-1/NOPB.B   | D            | SOIC         | 8    | 95  | 495    | 8      | 4064   | 3.05   |
| LM386N-1/NOPB     | P            | PDIP         | 8    | 40  | 502    | 14     | 11938  | 4.32   |
| LM386N-1/NOPB.B   | P            | PDIP         | 8    | 40  | 502    | 14     | 11938  | 4.32   |
| LM386N-3/NOPB     | P            | PDIP         | 8    | 40  | 502    | 14     | 11938  | 4.32   |
| LM386N-3/NOPB.B   | P            | PDIP         | 8    | 40  | 502    | 14     | 11938  | 4.32   |
| LM386N-3/NOPBG4   | P            | PDIP         | 8    | 40  | 502    | 14     | 11938  | 4.32   |
| LM386N-3/NOPBG4.B | P            | PDIP         | 8    | 40  | 502    | 14     | 11938  | 4.32   |
| LM386N-4/NOPB     | P            | PDIP         | 8    | 40  | 502    | 14     | 11938  | 4.32   |
| LM386N-4/NOPB.B   | P            | PDIP         | 8    | 40  | 502    | 14     | 11938  | 4.32   |
| LM386N-4/NOPBG4   | P            | PDIP         | 8    | 40  | 502    | 14     | 11938  | 4.32   |
| LM386N-4/NOPBG4.B | P            | PDIP         | 8    | 40  | 502    | 14     | 11938  | 4.32   |

**DGK0008A****PACKAGE OUTLINE****VSSOP - 1.1 mm max height**

SMALL OUTLINE PACKAGE



4214862/A 04/2023

**NOTES:**

PowerPAD is a trademark of Texas Instruments.

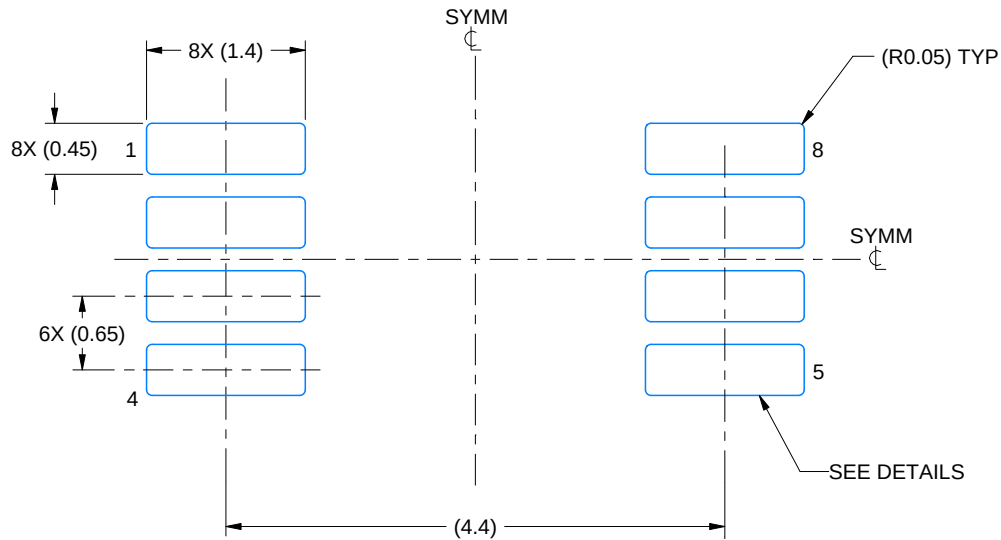
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-187.

# EXAMPLE BOARD LAYOUT

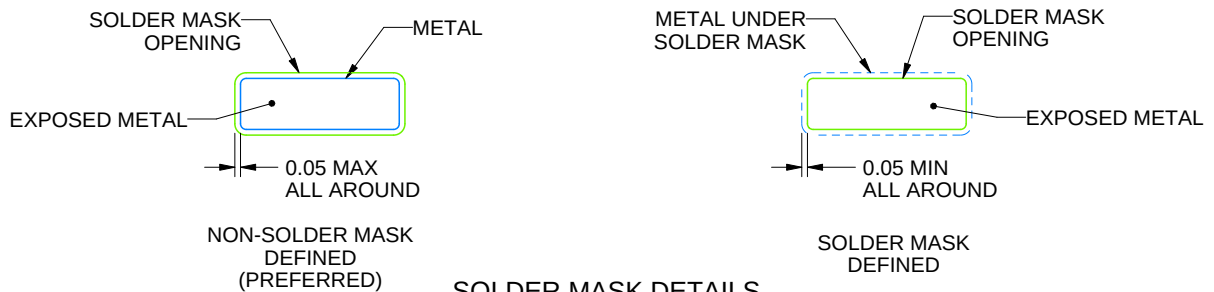
DGK0008A

<sup>TM</sup> VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE: 15X



SOLDER MASK DETAILS

4214862/A 04/2023

NOTES: (continued)

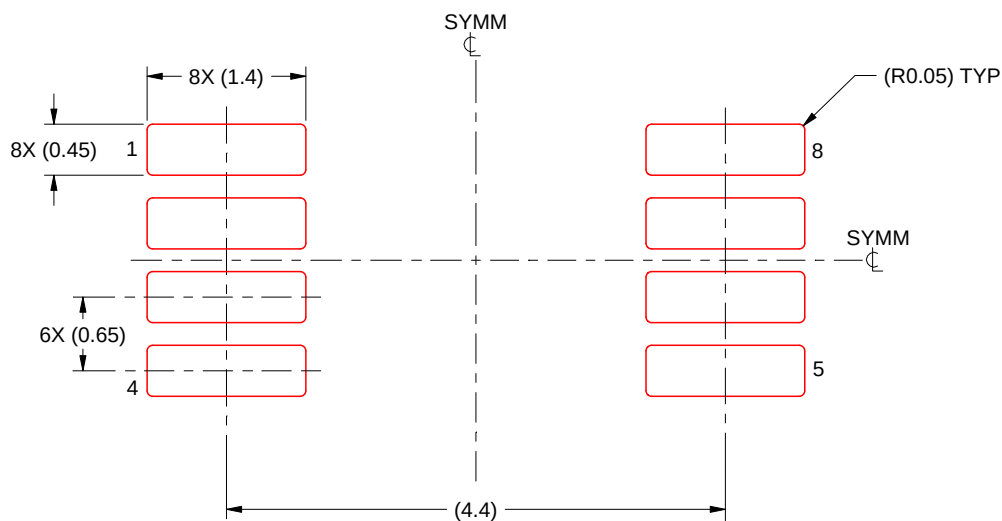
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
8. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.
9. Size of metal pad may vary due to creepage requirement.

## EXAMPLE STENCIL DESIGN

DGK0008A

<sup>TM</sup> VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE  
SCALE: 15X

4214862/A 04/2023

NOTES: (continued)

11. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
12. Board assembly site may have different recommendations for stencil design.



## PACKAGE OUTLINE

**SOIC - 1.75 mm max height**

SMALL OUTLINE INTEGRATED CIRCUIT



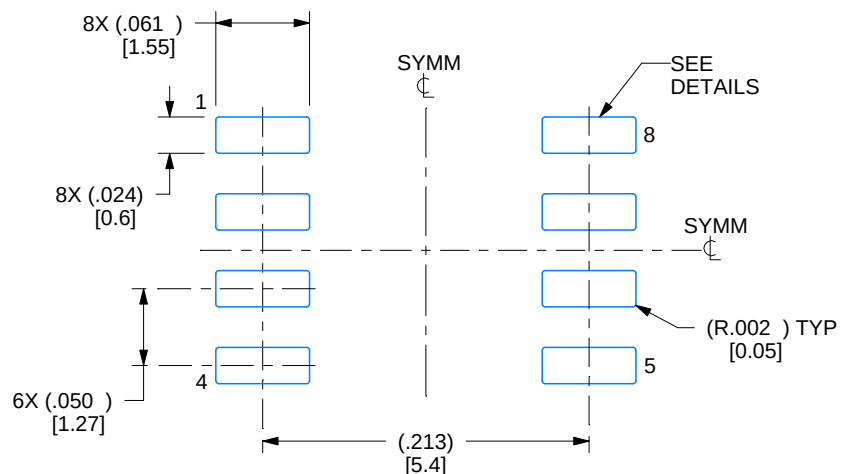
1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.

# EXAMPLE BOARD LAYOUT

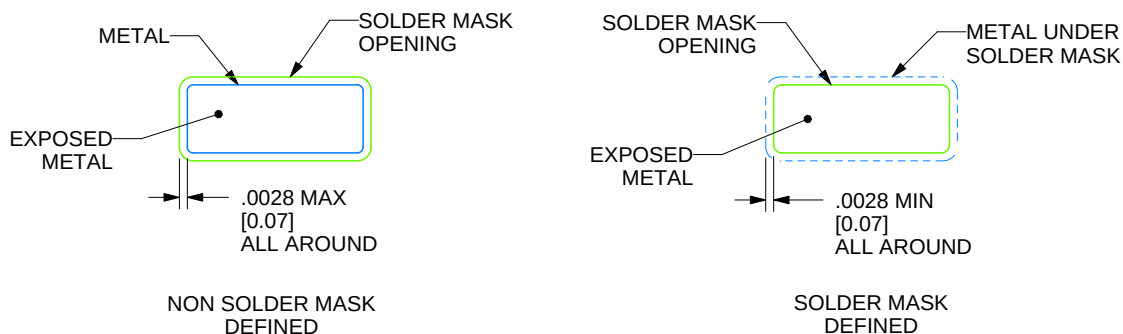
D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE:8X



SOLDER MASK DETAILS

4214825/C 02/2019

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

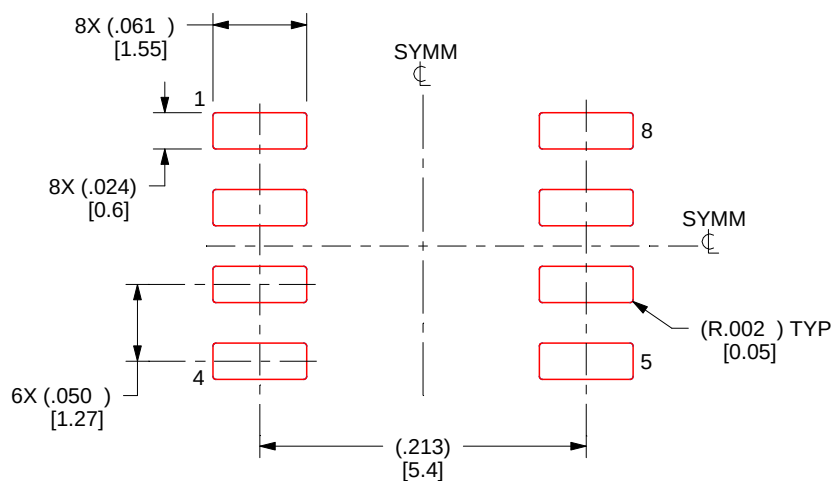
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE  
BASED ON .005 INCH [0.125 MM] THICK STENCIL  
SCALE:8X

4214825/C 02/2019

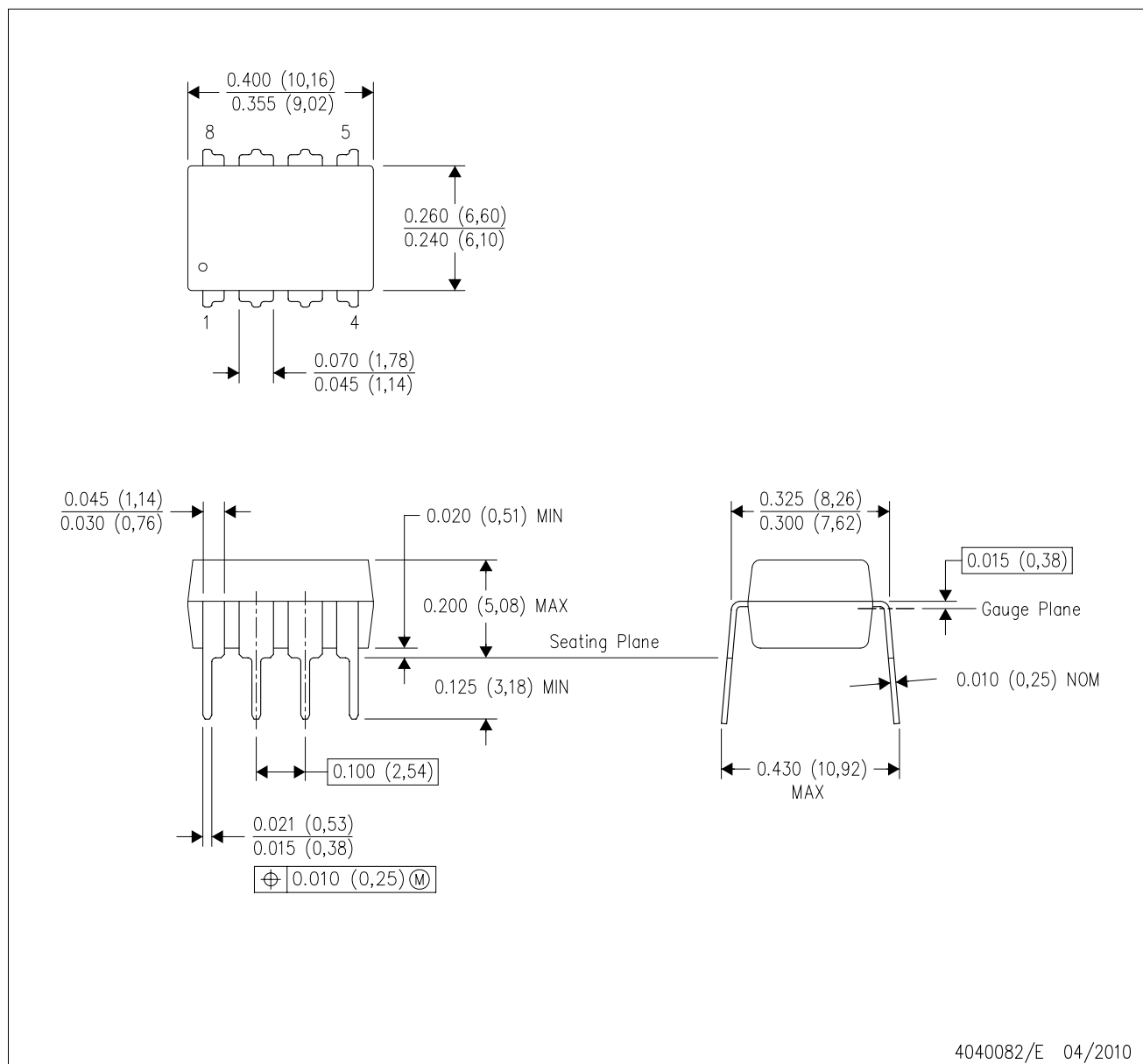
NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.



P (R-PDIP-T8)

PLASTIC DUAL-IN-LINE PACKAGE



- NOTES:
- A. All linear dimensions are in inches (millimeters).
  - B. This drawing is subject to change without notice.
  - C. Falls within JEDEC MS-001 variation BA.

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